Oil-Price Exposure as a Single-Factor Smart-Beta Strategy

Mock research note | Teaching and API integration example | 27 October 2026

Status: This is a synthetic methodology note, not a published academic study. No empirical equity-return tests, performance estimates, or statistically significant factor premia are asserted.

Abstract

This note proposes a single-factor equity strategy based on security-level sensitivity to changes in oil prices. Securities are assigned an oil-price exposure (OILP) estimated from historical returns and oil-price changes. A constrained long-short portfolio optimizer constructs positive-, zero-, and negative-OILP portfolios while controlling other modeled exposures. A separate timing signal selects which portfolio to hold. The associated monthly oil-level series from the supplied *OilPrices.csv* supports computation of oil changes and forward test targets; the series alone does not contain the security returns needed to estimate stock exposures or execute the strategy.

1. Economic hypothesis and interpretation

Oil-price innovations can affect firms differently through production costs, revenues, demand conditions, and discount-rate channels. We investigate whether an investable portfolio tilted toward firms with positive oil sensitivity performs differently from a portfolio with negative oil sensitivity when oil prices change. The proposed **single factor is oil-price sensitivity, named OILP**. A positive OILP portfolio has a positive portfolio beta to oil returns; a negative OILP portfolio has a negative beta. The paper does **not** claim that high-OILP equities unconditionally earn higher average returns.

For an illustrative conditional hypothesis, higher contemporaneous or correctly forecast *future* oil returns favor positive oil exposure, while lower oil returns favor negative exposure. This is a testable directional assumption, not a finding inferred from the CSV.

2. Source data and definition

The provided CSV contains columns **Date** and **Factor Level**. We treat Factor Level as a monthly oil-related level P(t), without assuming a specific crude benchmark, currency, inflation adjustment, or contract roll convention. Source coverage is January 2011–March 2024 (159 observations). Define monthly simple oil changes as **r_oil(t) = P(t) / P(t-1) - 1**. For interval forecasts, define **R_oil(t,T) = P(T) / P(t) - 1**.

3. Observed time series



Figure 1. Monthly Factor Level from the user-supplied OilPrices.csv; units and underlying oil benchmark are not specified in the dataset.

There are **159** monthly observations, no missing entries, an observed minimum of **16.55**, and maximum of **114.84**. The chart displays observed input data, not backtest results. The accompanying CSV is the machine-readable time series used for replication.

Date	Factor Level	Monthly return
2011-01-01	89.1705	—
2011-02-01	88.5784	-0.66%
2011-03-01	102.8565	+16.12%
2011-04-01	109.5325	+6.49%
2023-12-01	71.9000	-7.45%
2024-01-01	74.1524	+3.13%
2024-02-01	77.2490	+4.18%
2024-03-01	81.2780	+5.22%

4. Estimating equity-level OILP exposure

For each security i, estimate its oil sensitivity on a training sample using **r_i(t) = a_i + b_i,oil r_oil(t) + G_i X(t) + e_i(t)**, where X(t) represents other risk controls (e.g., market, credit, consumer, and employment proxies). The coefficient b_i,oil is the security-level **OILP** exposure. All explanatory series must be aligned in frequency and observation timing. The attached CSV alone is insufficient to estimate OILP; separate equity returns and control series are required.

A static estimate computed using any observation after a historical portfolio decision date is not point-in-time valid. For genuine historical tests, estimate exposures using only data available at each rebalance date, or label static exposure tests as illustrative.

5. Three smart-beta portfolio regimes

Let w_i be the signed portfolio weight on security i, and b_i,oil its estimated oil sensitivity. Portfolio oil exposure is **B_oil(w) = sum_i w_i b_i,oil**. Define three target portfolios:

Regime	Illustrative OILP target	Interpretation
Positive	+0.25	Benefit from positive oil shocks, ceteris paribus
Neutral	0.00	Remove modeled oil exposure
Negative	-0.10	Benefit from negative oil shocks, ceteris paribus

Targets are **illustrative user-defined inputs**, not inferred from the factor paper and not necessarily feasible in a particular universe. They correspond to the existing engine defaults (+0.25, 0.00, -0.10). The optimizer must reject infeasible allocations rather than silently relaxing constraints.

6. Portfolio construction methodology

On each chosen construction date, solve a constrained optimization across the available equity universe. In the existing Strategy Economics implementation, a trailing cross-sectional regression yields historical residual returns e_i. The optimizer seeks **max_w sum_i w_i e_i**, subject to target OILP exposure; targets for other included factors; preset gross long/short allocations; name-level weight bounds; and a nonnegative fitted residual objective. Exact feasibility depends on the investable universe and estimated exposures.

The selected equity portfolio must use the optimizer output with **Symbol** and **Weight** columns. Positive, neutral and negative portfolios are not necessarily long-only: they are **factor-exposure regimes** formed through long and short positions.

7. Timing signal and backtest design

At each rebalance date t, a factor-timing signal s(t) is mapped into a regime: positive when s(t) is at least threshold h, negative when s(t) is at most -h, and neutral otherwise. In the existing Monte Carlo exercise, signals of controlled predictive quality are simulated from standardized **future** oil returns to measure the hypothetical economic value of forecast information. This is an oracle-calibrated simulation, **not** a deployable forecast or a valid predictive trading backtest by itself.

Use monthly dates or explicitly defined holding intervals, compute realized stock returns only after the relevant decision timestamp, and report turnover and transaction costs separately if they are introduced. A genuine investable experiment would substitute a signal computed from information available at t.

8. Validation and evaluation

A valid factor-replication test should verify: (i) selected factor name equals **OILP**; (ii) positive/neutral/negative exposures match user targets within optimizer tolerance; (iii) there are no duplicate names or missing return histories; (iv) the rebalance-to-return alignment is correct; and (v) exposure estimates use appropriate historical information. Assess realized oil beta, factor-neutrality residuals, turnover, Sharpe, drawdowns, and sensitivity to exposure targets.

9. Connection to fund economics

After constructing daily return paths, compare management and performance fee cash flows under a fixed fee schedule and initial AUM. The marginal economic value of better oil forecasting is the discounted difference in fee cash flows between an improved-information strategy and a matched zero-information baseline, less development and implementation costs. This note supplies methodology and test inputs; it does **not** supply empirical cash-flow results.

10. Limitations

The dataset label does not identify which oil price or unit is represented. Simple level changes may differ from investable futures returns because of contract rolls, carry and financing. Oil shocks can coincide with broader macroeconomic regimes, and stock exposure estimates can be unstable. Static exposures and full-sample target normalization introduce look-ahead risk. The regression-residual objective may favor historical noise; robustness requires rolling estimation, point-in-time data, and true out-of-sample controls.

11. Explicit instructions for an LLM factor-extraction test

This note presents **one clearly separable equity factor**: OILP (sensitivity of security returns to oil-price movements). If an available factor dictionary contains **OILP: oil-price sensitivity of equity returns**, the factor should be classified as compatible. Factor direction should be interpreted carefully: the note proposes a **conditional positive relation** between oil returns and the returns to positive-OILP exposure, but makes **no unconditional expected-return-premium claim**. All numerical exposure targets are controlled by the implementation, not by the LLM.

Reproduction package

Attached data: OilPrices.csv (monthly Date, Factor Level). **Expected wrapper result:** selected_factor = OILP; paper_title = Oil-Price Exposure as a Single-Factor Smart-Beta Strategy; can_implement = true, conditional on compatible security-level OILP exposure data; cautions should mention benchmark/units, conditional-versus-unconditional direction, and point-in-time estimation.

This mock paper and its associated series are intended for software integration testing, not for citation as published research or investment advice.